

Gestation Guardian

A Mobile-First Maternal Health Monitoring Application

Abstract

Gestation Guardian is a mobile application built to help pregnant women keep track of their health in a simple and accessible way. The app lets mothers monitor things like blood pressure, weight, baby kicks, contractions, and other daily vitals, all from their phone. It also has a built-in risk scoring system that looks at the mother's health data and tells her whether things are looking normal or if she needs to see a doctor. The idea behind this project came from seeing how many women, especially in rural and semi-urban areas, do not have easy access to regular checkups or proper maternal health guidance.

Problem Statement

Maternal mortality is still a serious issue in many parts of the world, including India. One of the biggest reasons for this is that complications like preeclampsia, gestational diabetes, and abnormal fetal movement often go unnoticed until it is too late. Most women only get their vitals checked when they visit the hospital, which might be once a month or sometimes even less frequently. Between those visits, there is no proper way for them to track what is happening with their body or their baby.

Another major problem is awareness. A lot of first-time mothers do not know what symptoms to watch out for. Things like sudden swelling, persistent headaches, blurred vision, or reduced baby movements can be early warning signs of dangerous conditions, but many women either ignore them or do not realize they are serious. By the time they reach the hospital, the situation may have already become critical.

On top of that, most existing health apps are either too generic or too complicated. They are not designed specifically for pregnancy monitoring, and they definitely do not give any kind of risk assessment based on the data the user enters. There is a clear gap for a simple, focused app that helps mothers stay on top of their health without needing a medical degree to understand it.

Proposed Solution

Gestation Guardian addresses these problems by putting daily health monitoring directly into the hands of the mother. The app provides a clean dashboard that shows the current gestational week, trimester, and a progress tracker based on the last menstrual period. Women can log their blood pressure (systolic and diastolic), record daily vitals like weight, sleep hours, stress levels, body temperature, blood glucose, and urine protein levels, all within a few taps.

The app includes a fetal kick counter where mothers can run timed sessions to track how often the baby is moving, and a contraction timer that records the duration and gap between contractions. These are exactly the kind of things doctors ask about during checkups, so having this data logged and ready is genuinely useful.

What makes this app different is the risk scoring engine at its core. It takes all the data the user has entered,

both static factors like age, pregnancy history, existing conditions like hypertension or diabetes, and BMI, along with dynamic readings like the latest blood pressure, protein in urine, glucose levels, and symptoms reported that day such as severe headache, vision changes, upper abdominal pain, or sudden swelling. It then calculates a risk score and classifies it into a simple color-coded system: Green for low risk, Amber for moderate, and Red for high or critical. Based on the result, the app tells the mother exactly what to do, whether it is to continue normal monitoring, mention something at the next visit, contact the doctor today, or go to the hospital immediately.

There is also an AI chatbot built into the app that answers common pregnancy-related questions about diet, iron intake, sleep positions, kick counts, blood pressure, contractions, and nausea. It is not a replacement for a doctor, but it gives quick, reliable guidance when the mother has a doubt at odd hours.

The app also includes a medical history section, a care guide with trimester-specific advice, a health records hub that organizes all logged data in one place, a reminder system for appointments and medications, and user authentication so that data stays safe and personal. The data is stored both locally on the device and on the cloud, so nothing gets lost even if the phone goes offline.